

CS661: Project Proposal Report

GROUP-5

1. Introduction

Diabetes and hypertension are two of the most widespread chronic conditions worldwide and managing them effectively depends heavily on how well health systems deliver treatment and sustain control over the long term. The World Health Organization (WHO) tracks treatment and control coverage for these conditions across nearly every country in the world, broken down by sex, making this one of the most reliable and analytically rich slices of global health data available for visual analytics.

Our project aims to build an interactive, web-based visual analytics system that explores global inequality in diabetes and hypertension care, specifically, how treatment coverage and treatment effectiveness (control) vary across countries, WHO regions, World Bank income groups, sex, and time (1990-2022). Rather than working with the full WHO Health Inequality Data Repository, much of which is sparse, survey-dependent, or inconsistently disaggregated, we deliberately scoped the project to the subset of indicators that are complete, consistent, and reliable across the full set of countries and years.

This narrower but verified-clean scope lets us build a coherent, fully populated analytics system rather than one with large gaps in coverage and tell a focused story: where in the world are people being treated for diabetes/hypertension but not having it effectively controlled, and how has this gap evolved and differed by sex, region, and income level.

2. Data Source

The data source is the WHO Health Inequality Data Repository (Health Care System and Access module), published by the WHO Global Health Observatory (<https://www.who.int/data/inequality-monitor>). The full repository contains 197 countries, 1980-2023, and 33 indicators, but we audited every indicator for completeness and found that most are sparse: many financial-hardship and workforce indicators cover only a handful of countries per year, are missing national-level values entirely, or are only broken down by dimensions (e.g., wealth quintile, age group) that have very limited country coverage.

We therefore scoped the project to the six indicators below, all from the Non-Communicable Disease (NCD) family, which we verified have zero missing estimates, near-complete country coverage, and continuous multi-decade year coverage:

Indicator	Countries	Years	Missing estimates
Diabetes treatment coverage (age-standardized)	195	1990-2022	0
Diabetes treatment coverage (crude)	195	1990-2022	0
Hypertension treatment coverage (age-standardized)	195	1990-2019	0
Hypertension treatment coverage (crude)	194	1990-2019	0
Hypertension effective control coverage (age-standardized)	195	1990-2019	0
Hypertension effective control coverage (crude)	194	1990-2019	0

Each indicator also includes a Sex breakdown (Male/Female) for every country-year, and every row carries a complete WHO region (whoreg6) and an almost-complete World Bank 2025 income group (wbincome2025, ~98% populated,

missing only for 5 small territories). Together these six indicators account for roughly 71,000 of the dataset's ~100,000 rows, the dense, reliable core of the repository.

3. Specific Tasks

For this project, we will perform the following main tasks:

1. **Data Aggregation & Cleaning:** Extract and structure the six NCD indicators across all countries and years, verifying consistency of country naming (ISO3 codes) and indicator scales.
2. **Data Preprocessing:** Compute derived metrics such as the male-female coverage gap, and the treatment-to-control gap (the share of treated patients whose condition is not effectively controlled).
 - The Absolute Sex Equity Gap (ΔSex): $\text{Rate_Male} - \text{Rate_Female}$ to capture tracking inequities.
 - Systemic Care Pipeline Leakage (ΔLeak): $\text{Rate_Treatment_Crude} - \text{Rate_Control_Crude}$ to track clinical drop-off rates.
 - Demographic Variance Offset (ΔAge): $\text{Rate_Crude} - \text{Rate_Age-Standardized}$ to calculate how much an aging national profile skews raw disease burden.
3. **Data Visualization:** Develop interactive visual representations of treatment/control coverage across countries, regions, income groups, and time.
4. **Insights Generation:** Apply statistical methods (gap measures, trend analysis, age-standardized vs. crude rate comparison) to uncover patterns of inequality in chronic disease care.
5. **User Interaction:** Integrate filtering and querying features allowing users to select countries, regions, income groups, indicators, sex, and year ranges.

4. Visualization Tasks

We will implement the following key visual analytics tasks to extract meaningful insights from the dataset:

4.1 Global Coverage Map

An interactive world choropleth map showing the value of a selected indicator (diabetes or hypertension treatment/control coverage) by country, with a year slider to animate change across the full 1990-2022 range, fully supported since every indicator has near-complete country coverage in every year.

4.2 Sex Gap Analysis

Visualizes the male-female coverage gap for a selected indicator and country, highlighting where treatment and control disparities by sex are largest, using a dumbbell or diverging bar chart, filterable by indicator and country.

4.3 Temporal Trend Analysis

Tracks how treatment/control coverage has evolved over time for selected countries, WHO regions, or income groups, using multi-line time-series charts, continuous and gap-free across the full-time range.

4.4 Treatment-to-Control Cascade

Compares treatment coverage against control coverage for hypertension within the same country and year, exposing the “leaky pipeline” where patients are treated but not effectively controlled, a genuine analytical insight unique to having both indicators available together.

4.5 Regional and Income-Group Comparison

Compares average indicator values across the six WHO regions and four World Bank income groups using grouped bar charts and box plots, filterable by indicator and year.

4.6 Age-Standardized vs. Crude Rate Comparison

Since both age-standardized and crude versions exist for every indicator, this view shows how much of cross-country variation is explained by differences in population age structure versus genuine differences in care-system performance, using a paired scatter/slope chart.

5. Overall Solution

Our proposed solution will be a single-page, web-based interactive dashboard (not a long scrolling page) that lets users explore global diabetes and hypertension care inequality through linked visual analytics views. The system will feature a world map, trend charts, sex-gap and cascade charts, and regional/income comparisons, all connected through shared filters for country, WHO region, income group, indicator, sex, and year. Selections made in one view (e.g., clicking a country on the map) will update the other linked views, enabling users to drill down from a global overview to country-level detail. Because every indicator in scope is fully populated, the dashboard will have no missing-data gaps to work around, and can tell a complete, defensible story about where chronic disease care is failing to convert treatment into control.

6. Tech Stack

- **Backend:** Python (Dash / Flask), Pandas and NumPy for data processing and aggregation.
- **Frontend:** Plotly / Dash, with D3.js for custom interactive components (map, dumbbell, cascade charts).
- **Data Storage:** Preprocessed CSV/Parquet files, optionally loaded into SQLite for fast filtered queries.

7. Team Members & Responsibilities

Each member will contribute to different aspects of the project:

- **Data Processing, Cleaning & Analytics:** Stuti Singh (251083), Unnati Gangurde (231107)
- **Backend Development:** Talakola Maha Lakshmi (241082), Khasha Meganakshtra (240542)
- **Visualization Development:** Pyna Hema Lakshmi SatyaSree (240819), Sruja BG (240289), Annadevara Yashvanth Chary (240143)
- **Frontend Development:** Mavilla Bharadwaja (251110021), Singampalli Sri Varshith (251110074)

This project aims to leverage public health data analytics and visualization techniques to make global inequality in chronic disease care more transparent and accessible. By scoping to a verified-complete data subset, our system will deliver a fully reliable, gap-free analytics experience for public health students, researchers, and policy analysts exploring three decades of WHO diabetes and hypertension data.